ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** sour	Acid.
2	**(b)** hydrochloric acid	HCl.
3	**(d)** vinegar	Vinegar.
4	**(d)** soapy	Bases are bitter.
5	**(a)** sulphuric acid	It has sulphur.
6	**(c)** soap	Soap.
7	**(b)** acidic	Acetic acid.
8	**(d)** nitric acid	Nitric.
9	**(d)** water	Water.
10	**(b)** basic	Base.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A sour-tasting substance (1); Vinegar (1).	1+1
12	Turns red (1); Shows acid (1).	1+1
13	Citric acid (1); Lactic acid (1).	1+1
14	A bitter, soapy substance (1); Soap (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lemon (1); Vinegar (1); Curd (1).	1+1+1
16	Citric: lemon (1); Lactic: curd (1); Acetic: vinegar (1).	1+1+1
17	Turns red (1); Soap is basic (1); Yellow in acid (1).	1+1+1
	OR: Soap (1); Baking soda (1); Lime water (1).	1+1+1
18	Soap (1); Antacid (1); Cleaning (1).	1+1+1
	OR: Dark pink (1); Acid (1); Green in base (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Acid sour, base bitter (1); Acid: blue litmus red (1); Base: red litmus blue (1); Base feels soapy (1); Examples differ (1).	1+1+1+1+1
	OR: Litmus: red, blue (1); Turmeric: yellow, red (1); China rose: pink, green (1); Acid first (1); Base second (1).	1+1+1+1+1
20	Take red and blue litmus (1); Dip in the solution (1); See the colour change (1); Acid turns blue to red (1); Base turns red to blue (1).	1+1+1+1+1
	OR: Citric acid: lemon (1); Lactic acid: curd (1); Acetic acid: vinegar (1); Tartaric acid: tamarind (1); Formic acid: ants (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** sodium hydroxide	A base.
2	**(a)** soap solution	Base.
3	**(a)** neutral	Neutral.
4	**(b)** a base	Antacid.
5	**(d)** acid	Acid.
6	**(d)** acidic	Lactic acid.
7	**(b)** a base	Calcium hydroxide.
8	**(a)** turmeric	Turmeric.
9	**(d)** acidic	Tartaric acid.
10	**(a)** formic acid	Acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Turns blue (1); Shows base (1).	1+1
12	Soap (1); Baking soda (1).	1+1
13	Shows nature of a solution (1); Changes colour (1).	1+1
14	Neither acid nor base (1); Does not change litmus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water (1); Sugar solution (1); Salt solution (1).	1+1+1
16	HCl (1); H_2SO_4 (1); HNO_3 (1).	1+1+1
17	Salt (1); Water (1); Heat (1).	1+1+1
	OR: Litmus (1); Turmeric (1); China rose (1).	1+1+1
18	$NaOH$ (1); KOH (1); $Ca(OH)_2$ (1).	1+1+1
	OR: Stomach has too much acid (1); Tablet is a base (1); It gives relief (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cleaning (1); Pickles (1); Digestion (1); Batteries (1); Fertilisers (1).	1+1+1+1+1
	OR: Ant sting has formic acid (1); Apply baking soda (1); Indigestion due to excess acid (1); Take antacid (1); Base neutralises acid (1).	1+1+1+1+1
20	Soaps (1); Antacids (1); Whitewash (1); Cleaning (1); Fertilisers (1).	1+1+1+1+1
	OR: Wear gloves (1); Wear goggles (1); Do not taste (1); Pour acid into water (1); Wash spills (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** sugar	Sugar.
2	**(c)** red	Acid turns it red.
3	**(d)** a base	Neutralises.
4	**(d)** salt	Salt and water.
5	**(a)** blue	Base.
6	**(a)** bases	Neutralise stomach acid.
7	**(b)** common salt	Sodium chloride.
8	**(b)** natural indicator	Dye.
9	**(c)** salt and water	Products.
10	**(a)** acid	Acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ant bite has acid (1); Soda is a base (1).	1+1
12	Litmus (1); Turmeric (1).	1+1
13	Acid and base reaction (1); Gives salt and water (1).	1+1
14	Acetic acid (1); Sour taste (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sour in taste (1); Turn blue litmus red (1); Turn China rose dark pink (1).	1+1+1
16	Dip blue litmus (1); It turns red (1); So lemon is acidic (1).	1+1+1
17	Lime is a base (1); It neutralises soil acid (1); Improves crops (1).	1+1+1
	OR: Bitter (1); Soapy (1); Turn red litmus blue (1).	1+1+1
18	Dip red litmus (1); It turns blue (1); So soap is basic (1).	1+1+1
	OR: May be harmful (1); May be poison (1); Use indicators (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make paste of turmeric (1); Spread on paper (1); Dry (1); Cut strips (1); Use as indicator (1).	1+1+1+1+1
	OR: Soil may be acidic (1); Lime is a base (1); It neutralises acid (1); Compost improves soil (1); Crops grow better (1).	1+1+1+1+1
20	Acid + base (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); Salt is formed (1); Water is formed (1); No longer acidic or basic (1).	1+1+1+1+1
	OR: Lemon in food (1); Vinegar in pickles (1); Soap for washing (1); Antacid (1); Baking soda in cooking (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that acids turn red litmus blue; remember **acid: blue litmus becomes red**, and **base: red litmus becomes blue**.
- Expecting turmeric to change colour with an acid; turmeric changes (to red) only with a base.
- Forgetting that a neutral substance (sugar solution, water) gives *no change* with every indicator.
- Writing the products of neutralisation as only 'salt'; the products are **salt and water** (with heat).
- Mixing up the colours of China rose: acid gives *dark pink*, base gives *green*.
- Believing that every sour thing is dangerous; many edible substances (lemon, curd, vinegar) are weak acids, but laboratory acids must never be tasted.

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